Random Effects Models

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1 Overview

Objectives and overview

- This material is based Chapter 11–12 on the textbook "A First Course in Design and Analysis of Experiments" (Oehlert, 2000).
- Chapter outcomes include:
  - Students will be able to distinguish between a random effect and a fixed effect.
  - Students will be able to identify when random effects are appropriate.
  - Students will know how to implement random effects and mixed effects models.
  - Students will gain some familiarity with the `lme4` package in R.

Introduction

- So far, the treatment effects that we have been working with can be classified as *fixed-effects*: we wish to estimate the effect of specific (fixed) treatments.
- What if the treatments themselves are a random selection from a larger population?
- **Example:** A farmer has 100 pieces of equipment, and would like to know how the equipment used may impact crop-yield. Rather than testing all 100 pieces of equipment, they *randomly select* g pieces of equipment, and performs a CRD with these equipment.

Random Effects
A random effect is appropriate to model treatments that are a *random sample* from a population of potential treatments.
Random effects are useful if the focus of inference is on the *population*, not the individual treatment effects.

2 Models for Random Effects

Example: Carton factory

- We will build intuition of Random Effects models and designs using an example.

Carboard company
Suppose a company makes cardboard cartons for storing goods. They have 50 machines to make cartons, and want to understand the variation in quality for finished products.
They choose 10 machines at random from the 50, and make 40 cartons on each machine. The resulting cartons are tested for strength.

- This is a completely randomized design (CRD), with $g = 10$ treatments (the different machines), and $n_i = 40$, for a total of $N = 400$ EUs.
- The model we have been using for this type of data is:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \quad \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2).$$

- We are still going to use a very similar model. However, we are not as interested in the exact α_i for the 10 randomly chosen machines. Instead, we want to make inference on the larger population.
- We could proceed as usual, but that only tells us about the 10 specific machines that were randomly chosen, and we would get an entirely different set of machines if we repeated the experiment.
- Thus, we want to introduce a *random effect*: let the effect α_i represent a random draw from a population of possible effects:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \qquad \alpha_i \overset{iid}{\sim} N(0, \sigma^2_\alpha), \quad \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2).$$

- We no longer assume $\sum_i \alpha_i = 0$, rather $E[\alpha_i] = 0$.
- We have introduced a new parameter, σ^2_α, which describes the spread of the effects.
- *We assume that the two random components are independent.* That is, α_i and ϵ_{ij} are independent.
- If they are independent, the variance of the observed data is increased:

$$\text{Var}(y_{ij}) = \text{Var}(\mu + \alpha_i + \epsilon_{ij}) = \sigma^2_\alpha + \sigma^2.$$

- The parameters σ^2_α and σ^2 are called *variance components*.
- If observations belong to the same treatment group, then they now have non-zero correlation:

$$\text{Cor}(y_{ij}, y_{ik}) = \begin{cases} 1 & j = k \\ \sigma^2_\alpha/(\sigma^2_\alpha + \sigma^2) & j \neq k \end{cases}$$

- We call this within treatment group correlation the *intraclass* correlation.

Key summary

- In a random effects model, the parameters are μ, σ^2, and σ^2_α.
- We are not interested in making inferences about the specific α_i and ϵ_{ij}, which are treated as random variables, not parameters.
- Inference wise, we are now interested in confidence intervals and hypothesis tests regarding μ and σ^2.

3 Estimation: REML and LRT

Beyond ANOVA: Maximum Likelihood

- Classical random effects estimation relies on Expected Mean Squares (EMS) from ANOVA tables (See Oehlert (2000, Sections 11.3–11.6)).
- These approaches can be computed by hand, and would naturally extend our workflow with the `aov()` function and the `emmeans` package.
- Unfortunately, these approaches have their weaknesses for random (and mixed) effects models.
- Our models will remain the same, but how we analyze them will shift, keeping in-line with modern practices.
- For the type of models considered so far (fixed-effects), the two approaches are the same.
- Modern software relies on *Likelihood* methods.
- **Maximum Likelihood (ML)** finds the parameter estimates (mean and variances) that make the observed data most probable; confidence intervals are obtained using likelihood ratio tests.
- However, standard ML has a flaw when estimating variance components: it does not account for the degrees of freedom used to estimate the fixed effects (like the grand mean).
- This makes standard ML variance estimates *biased* (they underestimate the true variance).

  Example: For a normal distribution model, $y_{ij} \sim N(\mu_i, \sigma^2)$, the MLE of (μ, σ^2) is:

$$\left(\hat{\mu}_{MLE}, \hat{\sigma}^2_{MLE}\right) = \left(\bar{Y}_i, \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right).$$

  However, you may recall from you intro stats class that we usually don't estimate variance that way, instead we divide by n_1:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

  That's because the MLE for the variance component is biased low: on average, our estimate will be slightly smaller than the truth.

Restricted Maximum Likelihood (REML)

- To fix the bias in standard ML, modern software uses *Restricted Maximum Likelihood (REML).*
- REML mathematically removes the fixed effects from the likelihood equation before estimating the variance components.
- By accounting for the degrees of freedom lost in estimating fixed effects, REML provides *unbiased* estimates of σ^2 and σ^2_α.

  Example: building on the normally distributed model from before, the MLE for σ^2 was biased low:

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

  The REML estimate for this model is unbiased:

$$\hat{\sigma}^2_{REML} = s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- Additionally, REML handles unbalanced data effortlessly and prevents the impossible "negative variance" estimates that can sometimes occur when calculating variance using classical EMS methods.

Estimating variance components in R

- If we are primarily interested in the variance components σ^2_α, we will want to get the REML estimate for this component, as it is unbiased.
- We will shift from our `aov()` workflow and use the `lmer` function from the `lme4` package to do this in R:

```
# install.packages("lme4")
library(lme4)
lmer(response ~ 1 + (1 | randomEffect), data = data)
```

Example: Penicillin

- We will first consider the built-in `Penicillin` data in R, which contains the reduced diameter of bacterial growth, after penicillin application. Each measurement is taken on a specific `plate`, which is our random effect.

```
data("Penicillin")

output <- lmer(diameter ~ 1 + (1 | plate), data = Penicillin)
```

- What is the notation here doing? `1 + (1 | plate)`.
  - The model we are fitting is $y_{ij} = \mu + \alpha_i + \epsilon_{ij}$.
  - The `1` in the beginning is not strictly needed, but represents the grand mean μ.
  - The `(1 | plate)` represents the random effect. In `lme4`, the syntax `(A | B)` implies that B is the random effect grouping variable, and A is the effect that is random. For a simple random effects model, the mean (`1`) is always the thing that is random.
- Now let's look at the output.

```
summary(output)

Linear mixed model fit by REML ['lmerMod']
Formula: diameter ~ 1 + (1 | plate)
   Data: Penicillin

REML criterion at convergence: 613.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.41583 -0.55523  0.01212  0.59232  1.95241

Random effects:
 Groups   Name        Variance Std.Dev.
```

```
 plate    (Intercept) 0.09509  0.3084
 Residual             4.03333  2.0083
Number of obs: 144, groups:  plate, 24

Fixed effects:
            Estimate Std. Error t value
(Intercept)  22.9722     0.1788   128.5
```

Likelihood Ratio Tests (LRT)

- Instead of using exact F-tests derived from ratios of Mean Squares, modern mixed modeling often uses the *Likelihood Ratio Test (LRT)* for hypothesis testing.
- The LRT explicitly compares two competing models:
  1. A **Full Model** that includes the random effect.
  2. A **Reduced Model** (Null Model) that removes the random effect.
- The test measures how much the likelihood (fit) of the data drops when the random effect is removed.
- If the fit is significantly worse without the random effect, we reject $H_0 : \sigma_\alpha^2 = 0$. The test statistic follows a χ^2 distribution.

4 Estimating Variance Components

Modern Estimation in R

- Historically, variance components like σ_α^2 were estimated by solving systems of equations based on ANOVA tables and Expected Mean Squares (EMS).
- Modern statistical practice relies on *Restricted Maximum Likelihood (REML)*.
- REML provides unbiased estimates of variance components, handles unbalanced data seamlessly, and avoids the problem of negative variance estimates that can occur with ANOVA methods.
- In R, the standard package for fitting these models is `lme4`.

Implementing lme4
We use the `lmer()` function. The syntax requires us to specify both the fixed effects and the random effects.
For our model $y_{ij} = \mu + \alpha_i + \epsilon_{ij}$:

- The fixed grand mean (μ) is represented by `1`.
- The random machine effect (α_i) is represented by `(1 | machine)`, which tells R to fit a random intercept grouped by machine.

```r
# Install and load the lme4 package
# install.packages("lme4")
library(lme4)

# Fit the random effects model
```

```r
mod_carton <- lmer(strength ~ 1 + (1 | machine), data = carton_data)

# View the variance components
summary(mod_carton)
```

The Intraclass Correlation Coefficient (ICC)

- After fitting the model, we want to quantify how much of the total variance is driven by our random effect.
- We calculate the *Intraclass Correlation Coefficient (ICC)*:

$$ICC = \frac{\hat{\sigma}^2_\alpha}{\hat{\sigma}^2_\alpha + \hat{\sigma}^2}$$

- The ICC represents the proportion of total variance in the response that is accounted for by the grouping factor (the machines).
- An ICC of 0.40 means that 40% of the variation in carton strength is due to the machines, while the remaining 60% is random carton-to-carton error.

Calculating ICC in R

You can extract the variance components manually from the `summary()` output, or use the `performance` package for a direct calculation.

```r
# Install and load the performance package
# install.packages("performance")
library(performance)

# Calculate the ICC directly
icc(mod_carton)
```

Hypothesis Testing for Random Effects

- In a random effects model, the null hypothesis is that the population of treatments has zero variance:

$$H_0 : \sigma^2_\alpha = 0$$

- To test this in a modern framework, we use a *Likelihood Ratio Test (LRT)*.
- The LRT compares the fit of a model *with* the random effect against a model *without* the random effect.
- Note: When comparing models with different random effects structures via LRT, we must fit them using standard Maximum Likelihood (`REML = FALSE`) rather than REML.

```r
# 1. Fit the null model (fixed grand mean only, no random effect)
mod_null <- lm(strength ~ 1, data = carton_data)

# 2. Fit the random effects model using Maximum Likelihood
mod_alt <- lmer(strength ~ 1 + (1 | machine),
```

```
                data = carton_data,
                REML = FALSE)

# 3. Perform the Likelihood Ratio Test
anova(mod_null, mod_alt)
```

- The `anova()` function will return a χ^2 test statistic and a p-value.
- If the p-value is small (e.g., < 0.05), we reject H_0 and conclude that there is significant variation in quality among the population of machines.

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References

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5. 1, 4